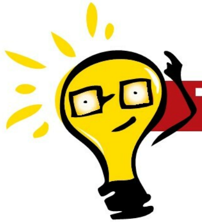

TIPS & TRICKS

Booting an ISO directly from the hard drive using GRUB 2

We often find ourselves in a situation in which we have an ISO image of Ubuntu on our hard disk and we need to test it by first running it. Try out this method for using the ISO image.

Create a GRUB menu entry by editing the */etc/grub.d/40_custom* file. Add the text given below just after the existing text in the file:

```
#gksu gedit /etc/grub.d/40_custom
```

Add the menu entry:

```
menuentry "Ubuntu 12.04.2 ISO"

{

set isofile="/home/<username>/Downloads/ubuntu-12.04.2-
desktop-amd64.iso"  #path of isofile

loopback loop (X,Y)$isofile

linux (loop)/casper/vmlinuz.efi boot=casper iso-scan/
filename=$isofile noprompt noeject

initrd (loop)/casper/initrd.lz

}
```

isofile variable is not required but simplifies the creation of multiple Ubuntu ISO menu entries.

The *loopback* line must reflect the actual location of the ISO file. In the example, the ISO file is stored in the user's *Downloads* folder. X is the drive number, starting with 0; Y is the partition number, starting with 1. *sda5* would be designated as *(hd0,5)* and *sdb1* would be *(hd1,1)*. Do not use (X,Y) in the menu entry but use something like *(hd0,5)*. Thus, it all depends on your system's configuration.

Save the file and update the GRUB 2 menu:

```
#sudo update-grub
```

Now reboot the system. The new menu entry will be added in the Grub boot option.

—***Kiran P S,***
pskirann@gmail.com

Playing around with arguments

While writing shell scripts, we often need to use different arguments passed along with the command. Here is a simple tip to display the argument of the last command.

Use *'!!:n'* to select the nth argument of the last command, and *'!$'* for the last argument.

```
dev@home$ echo a b c d

a b c d

dev@home$ echo !$

echo d

d

dev@home$ echo a b c d

a b c d

dev@home$ echo !!:3

echo c
c
```

—***Shivam Kotwalia,***
shivamkotwalia@gmail.com

Retrieving disk information from the command line

Want to know details of your hard disk even without physically touching it? Here are a few commands that will do the trick. I will use */dev/sda* as my disk device, for which I want the details.

```
smartctl -i /dev/sda
```

smartctl is a command line utility designed to perform